Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for the Airport Management System (AMS). The system is designed to support airport operations and improve the interaction between passengers, airport staff, and administrators.

This document specifies the system modules, actors, use cases, functional requirements, and non-functional requirements that describe how the system should operate.

The document serves as the foundation for later stages of the software engineering process, including system design, modeling, and development.

1.2 System Overview

Airports are complex environments where many operations must be coordinated simultaneously. These operations include passenger services, flight scheduling, baggage handling, security control, staff management, financial transactions, and airport resource management.

The Airport Management System (AMS) is an integrated software platform that manages these operations through a centralized digital system.

The system helps:

- Passengers access travel services
- Airport staff manage operations
- Administrators monitor system activities

The system improves efficiency, accuracy, and the overall airport experience.

1.3 System Scope

The Airport Management System (AMS) is designed to support and manage major airport operations through a centralized software platform. The system integrates multiple operational modules including passenger services, flight operations, baggage management, airport resources, security, finance, human resources, transportation services, and system administration. The system allows passengers to perform online check-in, track baggage, and receive flight notifications. Airport staff can manage operational activities such as assigning gates, handling baggage, and monitoring airport facilities. The system aims to improve operational efficiency, data accuracy, communication between departments, and the overall passenger experience at the airport.

1.4 Assumptions

- ✓ Users have access to an internet connection in order to interact with the system.
- ✓ Passengers must create user accounts before booking tickets or performing online check-in.
- ✓ Airport staff are properly trained to operate the system modules.
- ✓ The system will be accessed primarily through web-based interfaces.
- ✓ The system database will store passenger, flight, baggage, and operational information.

1.5 Constraints

- ✓ The system must comply with aviation security and data protection regulations.
- ✓ The system must operate using a centralized relational database.
- ✓ The system must be accessible through modern web browsers.
- ✓ The system must ensure data privacy and protection of passenger information.
- ✓ The system must integrate with existing airport information systems where required.

1.6. Future Enhancements

Future versions of the Airport Management System may include additional advanced features such as:

- ✓ Mobile application support for passengers and airport staff.
- ✓ Biometric passenger identification systems.
- ✓ AI-based flight delay prediction.
- ✓ Automated baggage tracking using IoT technology.
- ✓ Real-time airport navigation for passengers.
- ✓ Integration with airline loyalty and reward programs.

2. System Users (Actors)

The system supports multiple types of users who interact with it in different ways.

Passenger - Passengers are the primary users of the system. They interact with the system to manage travel-related services such as checking flight status, performing check-in, and tracking baggage.

Visitor - Visitors are users who access the system. They may want to check flight information or explore airport facilities.

Airport Staff - Airport staff handle operational tasks such as baggage processing, gate assignment, and facility management.

Security Officer - Security officers ensure airport safety by verifying passenger identities, monitoring security checkpoints, and managing restricted areas.

Customer Support Agent - Customer support agents assist passengers with inquiries, complaints, and service requests.

Finance Staff - Finance staff manage financial transactions, billing processes, and financial reporting related to airport services.

Human Resource Staff - Human resource staff manage employee records, staff scheduling, and role assignments.

System Administrator - System administrators manage the overall system operation including user accounts, system permissions, and maintenance.

3. Major System Modules – User Characteristics

The Airport Management System is divided into several functional modules. Each module manages a specific group of airport operations.

3.1 Passenger Service Management Module

This module manages services directly related to passengers.

Main Responsibilities

- Online check-in
- Boarding pass generation
- Flight notifications
- Passenger support

Actors

- Passenger
- Customer Support Agent

Example Use Cases

- Perform Online Check-In
- Download Boarding Pass
- Receive Notifications

3.2 *Flight Operations Management Module*

This module manages flight scheduling and flight operations.

Main Responsibilities

- Flight status updates
- Gate assignment
- Flight delay management

Actors

- Airport Staff

Example Use Cases

- Update Flight Status
- Assign Gate
- Manage Delayed Flights

3.3 Baggage Management Module

This module manages passenger baggage tracking and handling.

Main Responsibilities

- Baggage registration
- Baggage tracking
- Baggage loading
- Lost baggage reporting
- Baggage claim management

Actors

- Airport Staff
- Passenger

Example Use Cases

- Register Baggage
- Track Baggage
- Report Lost Baggage
- Update Baggage Location

3.4 Security Management Module

This module ensures airport safety and security.

Main Responsibilities

- Identity verification
- Security checkpoint monitoring
- Security incident reporting
- Restricted area control

Actors

- Security Officer

Example Use Cases

- Verify Passenger Identity
- Monitor Security Checkpoints
- Record Security Incidents

3.5 Airport Resource Management Module

This module manages airport infrastructure and resources.

Main Responsibilities

- Gate allocation
- Runway scheduling
- Terminal management
- Equipment monitoring

Actors

- Airport Staff

Example Use Cases

- Assign Gates
- Schedule Runway Usage
- Monitor Terminal Capacity

3.6 Finance and Billing Module

This module manages financial transactions related to airport services.

Main Responsibilities

- Service billing
- Financial reporting
- Transaction tracking

Actors

- Passenger
- Finance Staff

Example Use Cases

- Process Ticket Payment
- Record Payment Transactions
- Generate Financial Reports

3.7 Human Resource Management Module

This module manages airport employees and staff operations.

Main Responsibilities

- Employee records
- Staff scheduling
- Role assignment
- Staff management

Actors

- HR Staff
- System Administrator

Example Use Cases

- Register Employee
- Assign Staff Roles
- Manage Staff Schedules

3.8 Customer Support Module

This module handles passenger support services.

Main Responsibilities

- Inquiry management
- Complaint management
- Assistance services
- Feedback handling

Actors

- Passenger
- Customer Support Agent

Example Use Cases

- Submit Complaint
- Request Assistance
- Track Support Requests

3.9 Transportation and Parking Module

This module manages airport transportation services.

Main Responsibilities

- Parking management
- Taxi services
- Shuttle services
- Transportation information

Actors

- Passenger
- Visitor

Example Use Cases

- View Parking Availability
- Book Parking Space
- View Transportation Options

3.10 System Administration Module

This module manages the system infrastructure.

Main Responsibilities

- User account management
- System monitoring
- Permission control
- System maintenance

Actors

- System Administrator

Example Use Cases

- Manage User Accounts
- Monitor System Activity
- Update System Settings

4. Requirements

4.1 Functional Requirements organized by System Modules

4.1.1 Passenger Service Management

These requirements describe the services that the system provides directly to passengers.

1. Passenger views flight schedules
2. Passenger uploads travel documents
3. Passenger performs online check-in
4. Passenger downloads boarding pass
5. Passenger checks assigned gate information
6. Passenger tracks baggage status
7. Passenger receives flight notifications

4.1.2 Flight Operations Management

These requirements manage flight scheduling and airline operations.

8. Airport staff updates flight status
9. Airport staff manages flight delays
10. Airport staff updates departure times
11. Airport staff updates arrival times
12. Airport staff coordinates boarding processes
13. Airport staff monitors flight operations

4.1.3 Baggage Management System

These requirements handle passenger baggage processing and tracking.

14. Airport staff registers passenger baggage
15. Airport staff tracks baggage movement
16. Airport staff records baggage loading

17. Airport staff updates baggage location
18. Airport staff reports lost baggage
19. Airport staff manages baggage claims
20. Passenger reports delayed baggage
21. Airport staff schedules baggage delivery

4.1.4 Security Management System

These requirements support airport security and safety operations.

22. Security officer verifies passenger identity
23. Security officer monitors security checkpoints
24. Security officer records security incidents
25. Security officer manages restricted area access
26. Security officer verifies boarding pass at security checkpoints
27. Security officer manages emergency security alerts

4.1.5 Airport Resource Management

These requirements manage airport infrastructure and operational resources.

28. Airport staff assigns gates to flights
29. Airport staff schedules runway usage
30. Airport staff monitors terminal occupancy
31. Airport staff manages airport facility resources

4.1.6 Customer Support Management

These requirements handle passenger assistance and customer service operations.

32. Customer support agent responds to inquiries
33. Customer support agent tracks service requests

4.1.7 Finance and Billing Management

These requirements manage financial transactions and billing processes.

34. Finance staff generates billing reports
35. Finance staff processes service charges
36. Finance staff generates financial transaction reports

4.1.8 Human Resource Management

These requirements support airport staff management.

37. HR staff registers employee records
38. HR staff manages employee schedules
39. HR staff assigns staff roles

4.1.9 System Administration

These requirements manage the system infrastructure and user control.

40. System administrator manages user accounts
41. System administrator updates airport information
42. System administrator monitors system usage
43. System administrator manages system permissions
44. System administrator performs system maintenance
45. System administrator generates system activity reports

4.2 Non-Functional Requirements (50)

Non-functional requirements describe how the system (Airport Management System) should perform.

4.2.1 Product Requirements

Product requirements describe how the system behaves and its quality attributes (usability, performance, security, availability).

4.2.1.1 Usability Requirements

1. The system user interface must comply with WCAG 2.1 Level AA accessibility standards to accommodate users with disabilities.
2. The system must allow easy navigation.
3. Users must be able to complete core tasks (e.g., checking flight status, finding a gate) within a maximum of 3 clicks from the homepage.
4. The system must provide clear instructions.
5. The system must display helpful error messages.
6. The system must provide user help documentation.
7. The interface must be visually consistent.
8. The system must be accessible to first-time users.

4.2.1.2 Performance Requirements

9. The system must load pages within three seconds.
10. The system must process searches within two seconds.
11. The system must support at least 1000 concurrent users.
12. Payment processing must complete within five seconds.
13. The system must support at least 1500 simultaneous users during peak travel periods.
14. The system must optimize database queries.
15. The system must maintain stable performance under high workload conditions.
16. The system must process requests without noticeable delays.

4.2.1.3 Availability

17. The system must operate 24/7.
18. System downtime must not exceed 1% of total operational time.

19. Maintenance must occur during low-traffic hours.

4.2.1.4 Security

20. The system must require user authentication.
21. Passwords must be encrypted.
22. The system must protect personal data.
23. The system must prevent unauthorized access.
24. Sessions must expire after inactivity.
25. Login attempts must be logged.
26. Data transmission must use secure communication protocols.
27. Sensitive information must be protected.
28. The system must implement Role-Based Access Control (RBAC) to ensure staff can only access modules relevant to their job title.

4.2.1.5 Portability

29. The web application must be fully functional on the latest two major versions of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.
30. The mobile web interface must support responsive design optimized for iOS 15+ and Android 12+.
31. The system must operate smoothly on-screen resolutions ranging from 360x800 pixels (mobile) to 1920x1080 pixels (desktop).

4.2.2 Organizational Requirements

These requirements relate to how the system is developed, maintained, and structured.

Maintainability

32. The system must have modular code.
33. The system must be well documented.
34. Updates must be easy to apply.
35. Bugs must be easy to identify and fix.
36. Developers must be able to add new features easily.

Reliability

37. The system must maintain at least 99% system uptime.
38. The system must recover from minor failures.
39. The system must maintain data integrity during all transactions.

40. The system must prevent data corruption.
41. The system must perform automated daily incremental backups and weekly full backups, retaining data for a minimum of 30 days.
42. System errors must be logged.
43. The system must support future system expansion.
44. The system must handle large volumes of operational and passenger data efficiently

Data Management

45. User data must be stored securely.
46. Flight data must remain accurate.
47. The system must maintain data consistency across all system modules.

4.2.3 External Requirements

These requirements concern interaction with external environments (devices, systems, platforms).

Compatibility

48. The system must comply with international and national government laws.
49. The system must coordinate with other airport systems and air traffic control.
50. The system must comply with Airport Collaborative Decision Making (A-CDM) standards.

5. User Scenarios / Use Cases

Use Case 1: Passenger Performs Online Check-in

Actor: Passenger

Description:

A passenger verifies their travel documents and generates a boarding pass for an upcoming scheduled flight.

Steps:

1. Passenger logs into the system.
2. Passenger inputs their flight reference number.
3. Passenger verifies travel document information
4. Passenger confirms check-in

Outcome:

The system generates a boarding pass and updates the passenger's check-in status.

6. Conclusion

The Airport Management System integrates multiple airport operations into a unified digital platform. The requirements described in this document define the system's expected behavior, operational capabilities, and quality attributes.

These requirements will guide future development phases including system modeling, architectural design, database design, and implementation.